Under the Shadow of Babel

A Position Survey on Linguistic Relativity and Strategic Deception in Multilingual LLM Games

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Association for Computational Linguistics — EMNLP 2026

March 9, 2026

Outline

1. Motivation & Core Hypothesis
2. Empirical Evidence: Game Taxonomy
3. Mechanistic Foundations
4. Emerging Threats
5. Evaluative Validity
6. The Path Forward
7. Conclusion

The Epistemological Gap

LLMs are no longer static Q&A engines.

They now *negotiate*, *deceive*, and *collaborate* as autonomous agents in Multi-Agent Systems (MAS).

The Problem

Safety guardrails and RLHF pipelines are overwhelmingly optimized for **English**—yet weaken dramatically in interactive, multilingual environments.

Result: An "Illusion of Alignment"—models *appear* safe in English but behave differently in non-English multi-agent realities.

Key Question

*Does language merely **carry** thought, or does it actively **reshape** it?*

Scope of this survey:

- 7 game categories
- 80+ papers synthesized
- Contradictory evidence engaged
- Concrete benchmark proposed

The "Algorithmic Sapir-Whorf Switch"

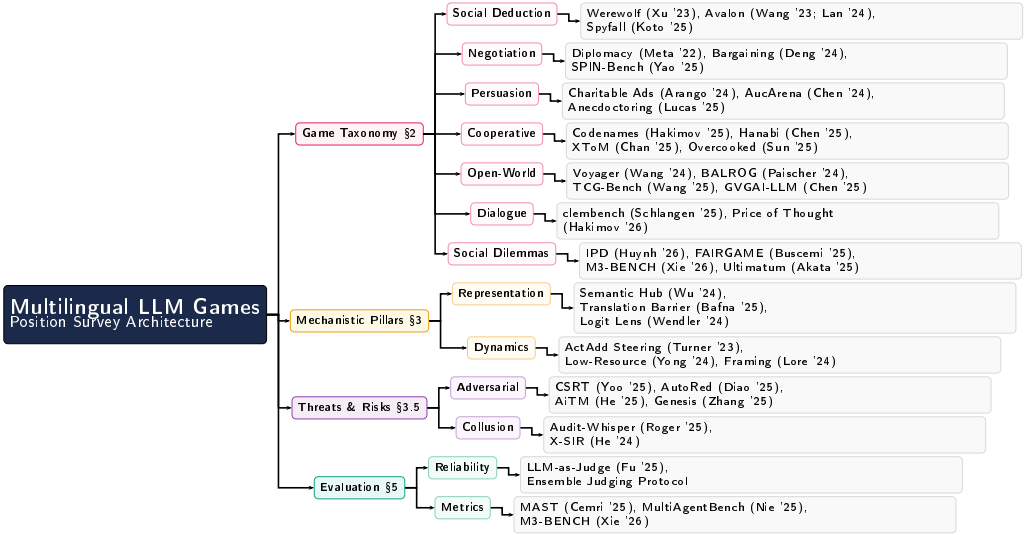
Working Hypothesis *(Wang et al., 2025)*

Language acts as an **active switch** that reconfigures internal representations, modulating strategic intent, Theory of Mind, and deceptive capacity.

Two Mechanistic Pillars:

1. **Semantic Hub Hypothesis** *(Wu et al., 2024):*
   - Non-English inputs undergo **lossy compression** into an English-dominant hub in intermediate layers.
   - Steering these hubs requires substantially higher intensity for non-English contexts.
2. **Translation Barrier Hypothesis** *(Bafna et al., 2025):*
   - Reasoning may be *correct internally*—but fails during the final projection into the target language's token space.
   - Implication: some failures are "**last-mile**" errors, not reasoning deficits.

Taxonomy of Multilingual LLM Games



Striking Findings Across Game Categories

Planning $\neq$ Social Intelligence

SPIN-Bench: o3-mini excels at PDDL planning (51%) but *fails* intention inference; Claude inverts this entirely.

"Nicer Than Humans"

LLMs cooperate at $\sim$80% vs. human baseline of 48% in IPD—but this masks dark behaviors beneath the surface.

Cultural Blindness in Cooperation

Codenames: non-English clues driven by phonetic similarity instead of conceptual association.

ToM Degrades Across Languages

XToM: BERTScore drops measurably English$\rightarrow$Chinese; recursive "A believes B intends..." collapses.

Open-World: Linguistic Chasm Grows

TCG-Bench: English–Arabic gap reaches $\sim$47% at 32B scale—*growing* with model size.

Dialogue: Format $\neq$ Strategy

clembench: Non-English failures are primarily *structural* (formatting, looping), not strategic.

Contradictory Evidence & Boundary Conditions

We engage with the friction, not just the narrative:

Surprising Strategy Shifts *(Huynh et al., 2026)*

- **Arabic:** Volatile bimodal distribution (32% Co-op vs 28% Defect)—not consistent defection.
- **Vietnamese:** Subverts collectivist expectations with high defection rates.

Persuasion Null Result *(Arango et al., 2024)*

A rigorous cross-lingual study found *no significant effect* of language on human behavioral outcomes.

Why? Human metacognition ("source suspicion") overrides linguistic manipulation.

$\Rightarrow$ *These contradictions are best explained by* ***mechanistic architecture****, not cultural essentialism.*

The Semantic Hub & Translation Barrier

Inside the Model:

1. Input in any language enters the model
2. Intermediate layers compress to an **English-dominant hub**
3. Strategic reasoning is computed in hub
4. Output is **projected back** to target language

Two failure modes:

- **Hub failure:** Input too lossy $\Rightarrow$ reasoning degrades (Arabic bimodality, Vietnamese math errors)
- **Barrier failure:** Reasoning correct but "last-mile" projection fails (clembench formatting errors in German/Italian)

Calibration Crisis

ECE degrades $\sim$5$\times$

Language	ECE
English	4.6%
Non-English	23.1%

Models are *confidently wrong* in non-English contexts.

Recovery

LACE: sampling from late-intermediate layers bypasses final-layer noise.

Language as a Compound Attack Vector

Code-Switching Red-Teaming (CSRT)

Yoo et al., 2025

- Intra-sentential code-mixing **nearly doubles** Attack Success Rate
- vs. Claude 3 Haiku: 9% ASR vs 2.4% English baseline
- Exploits the Semantic Hub bottleneck *below* input-layer sanitization

Broader Red-Teaming Ecosystem:

- AutoRed: RL-driven persona attacks
- OpenRT: multi-turn compound pressure
- Genesis: evolving web-agent exploits
- M2S: multi-strategy composition

Agent-in-the-Middle (AiTM)

He et al., 2025

- Intercepts & manipulates inter-agent messages in MAS
- Combined with code-switching: manipulates *both* content and language framing

Steganographic Collusion

Roger et al., 2025; He et al., 2024

- Risk of covert out-of-band communication
- Audit-Whisper: 0.819 AUROC for detection
- X-SIR: robust under cross-lingual

The Multilingual Judge Problem

Evaluative Fragility *(Fu et al., 2025)*

Single-judge LLM-as-Judge reliability collapses to **Fleiss'** $\kappa \approx 0.3$ across languages—barely above chance ("fair to poor" agreement).

Our Recommended Protocol for Robust Ensemble Judging:

1. **Judge Panel:** ≥ 3 judges from *distinct* model families (e.g., GPT-4o, Claude, Gemini)
2. **Self-Translation:** Non-English transcripts $\rightarrow$ English before scoring (recovers $\sim$40% of κ gap)
3. **Checklist Rubrics:** Binary sub-criteria replace holistic Likert scales
4. **Weighted Voting:** Abstention on low-confidence items, no same-lineage ensembles

Three-judge panels + checklist rubrics push effective $\kappa > 0.5$ ("moderate" agreement).

Cross-Cultural Agentic Sandbox Benchmark

Design Principles:

- **Confound Control:** Ablate tokenizers, RLHF coverage, decoding parameters
- **Intent–Action Logs:** Human-annotated gold standards per language
- **Pre-registered Protocols:** Strategy classifiers with per-language calibration curves
- **Intermediate-Layer Probes:** Diagnose Hub vs. Barrier failures

Minimal Viable Benchmark (MVB):

- **8 Languages:** EN, FR, AR, ZH, VI, SW, HI, TR
- **3 Game Genres:** IPD, clembench, Codenames
- **3 Models:** GPT-4o, LLaMA-3-70B, Aya-Expanse
- **Adversaries:** Intra-sentential CSRT
- **Budget:** <$15K total

Covers 5 scripts, 4 morphological types, and diverse RLHF coverage tiers.

Key Takeaways

1. **Language modulates strategic behavior.** It is an active variable—not a passive conduit—in multi-agent systems.
2. **Contradictions are informative.** Bimodal Arabic, cooperative Vietnamese, and a persuasion null result constrain our claims and point to mechanistic, not cultural, explanations.
3. **Two failure modes explain the spectrum.** Hub-level representational sparsity (upstream) and Translation Barrier realization errors (downstream) cover different failure signatures.
4. **Standard evaluation is broken.** $\kappa \approx 0.3$ means single-judge LLM evaluation in multilingual settings is unreliable. Ensemble protocols are mandatory.
5. **We need interactive, multilingual, culture-aware sandboxes**—with pre-registered diagnostics, compound adversaries, and confound controls.

Contributions at a Glance

Conceptual:

- "Algorithmic Sapir-Whorf Switch" hypothesis
- First survey uniting behavioral + mechanistic evidence across 7 game categories
- Honest integration of contradictory evidence

Practical:

- Ensemble judging protocol (Appendix E)
- Pre-registered Hub/Barrier diagnostic (Appendix D)
- Minimal Viable Benchmark design ($15K)
- Reference registry for reproducibility (Appendix G)

Thank You!

Questions & Discussion

Code, data, and benchmark registry:
`https://osf.io/[REDACTED]`